

World Development Indicators Analysis (2022)

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This report analyses a sample of the World Development Indicators (WDI) dataset for 2022. The data source is the World Bank WDI database (World Bank 2022).

Exploratory Data Analysis

We focus on three key indicators for 2022: GDP per capita, life expectancy, and inflation rate. These variables capture economic development, health outcomes, and macroeconomic stability.

```
df[["gdp_per_capita", "life_expectancy", "inflation_rate"]].describe()
```

	gdp_per_capita	life_expectancy	inflation_rate
count	203.000000	209.000000	169.000000
mean	20345.707649	72.416519	12.493936
std	31308.942225	7.713322	19.682433
min	259.025031	52.997000	-6.687321
25%	2570.563284	66.782000	5.518129
50%	7587.588173	73.514634	7.967574
75%	25982.630050	78.475000	11.665567
max	240862.182448	85.377000	171.205491

The (**eda-summary?**) reveal substantial cross-country variation in economic conditions. GDP per capita has a mean of about 20,346 USD but a much lower median of 7,588 USD, indicating strong right-skewness driven by high-income outliers and large dispersion.

Life expectancy is less variable, with a mean of 72.4 years and a relatively small standard deviation, suggesting more moderate differences across countries.

Inflation exhibits the greatest volatility. Although the median rate is around 8 percent, extreme maximum values (above 170 percent) inflate both the mean and standard deviation, confirming the presence of severe outliers and macroeconomic instability in some countries.

Visualizing Key Relationships

The following plots illustrate the patterns identified in the exploratory analysis.

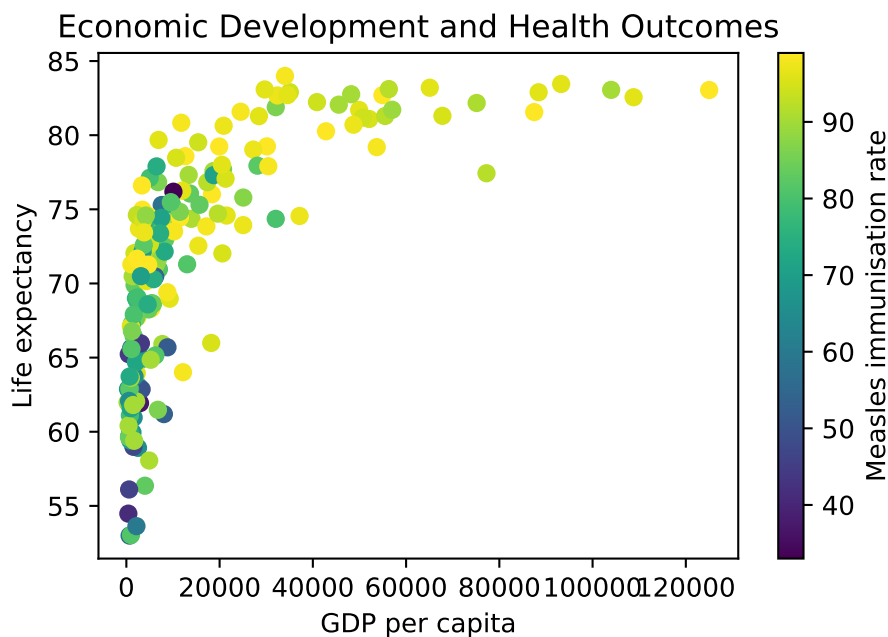


Figure 1: GDP per capita vs Life Expectancy, colored by measles immunisation rate (2022).

Figure 1 visually confirms the positive association between income and life expectancy identified in the correlation analysis.

Figure 2 illustrates the right-skewed distribution of inflation, consistent with the presence of extreme outliers observed in the summary statistics.

Summary of Key Indicators

```
df[["gdp_per_capita", "life_expectancy", "inflation_rate"]] \
  .agg(["mean", "median", "std"]) \
  .T \
  .round(2) \
  .set_axis(["Mean", "Median", "Std Dev"], axis=1)
```

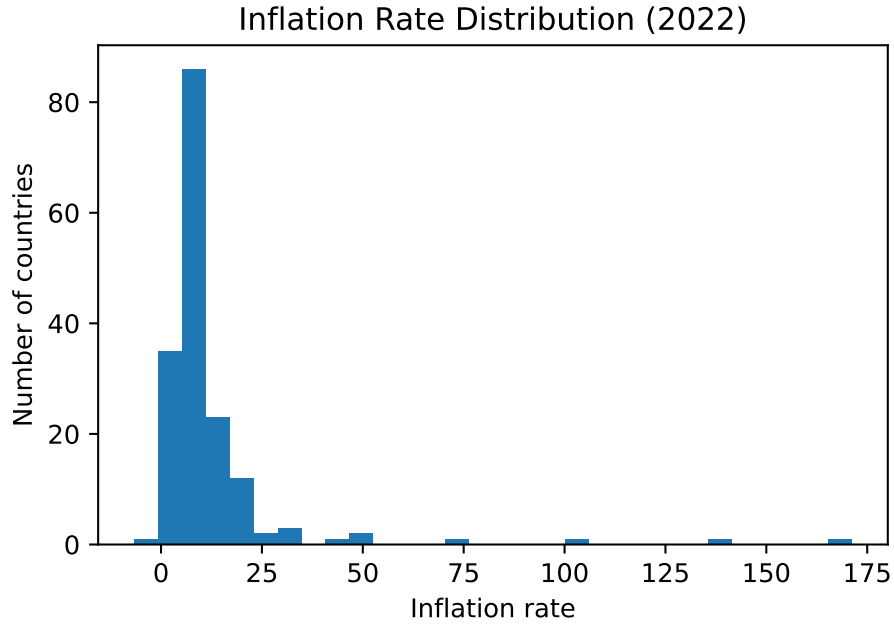


Figure 2: Distribution of inflation rates across countries (2022).

Table 2: Key statistics for selected indicators (2022).

	Mean	Median	Std Dev
gdp_per_capita	20345.71	7587.59	31308.94
life_expectancy	72.42	73.51	7.71
inflation_rate	12.49	7.97	19.68

Table 2 reports the mean, median, and standard deviation for GDP per capita, life expectancy, and inflation rate in 2022. GDP per capita exhibits substantial dispersion, reflecting large differences in income levels across countries. Life expectancy shows comparatively lower variability, while inflation displays greater volatility, consistent with the presence of extreme values observed earlier. Overall, the table reinforces the unequal distribution of economic and macroeconomic conditions worldwide.

The positive relationship between income and life expectancy is consistent with classic findings in the development literature (Preston 1975).

References

- Preston, Samuel H. 1975. “The Changing Relation Between Mortality and Level of Economic Development.” *Population Studies* 29 (2): 231–48.
- World Bank. 2022. “World Development Indicators.” <https://data.worldbank.org/indicator>.